

SPS_Framework_Supporting_Evidence

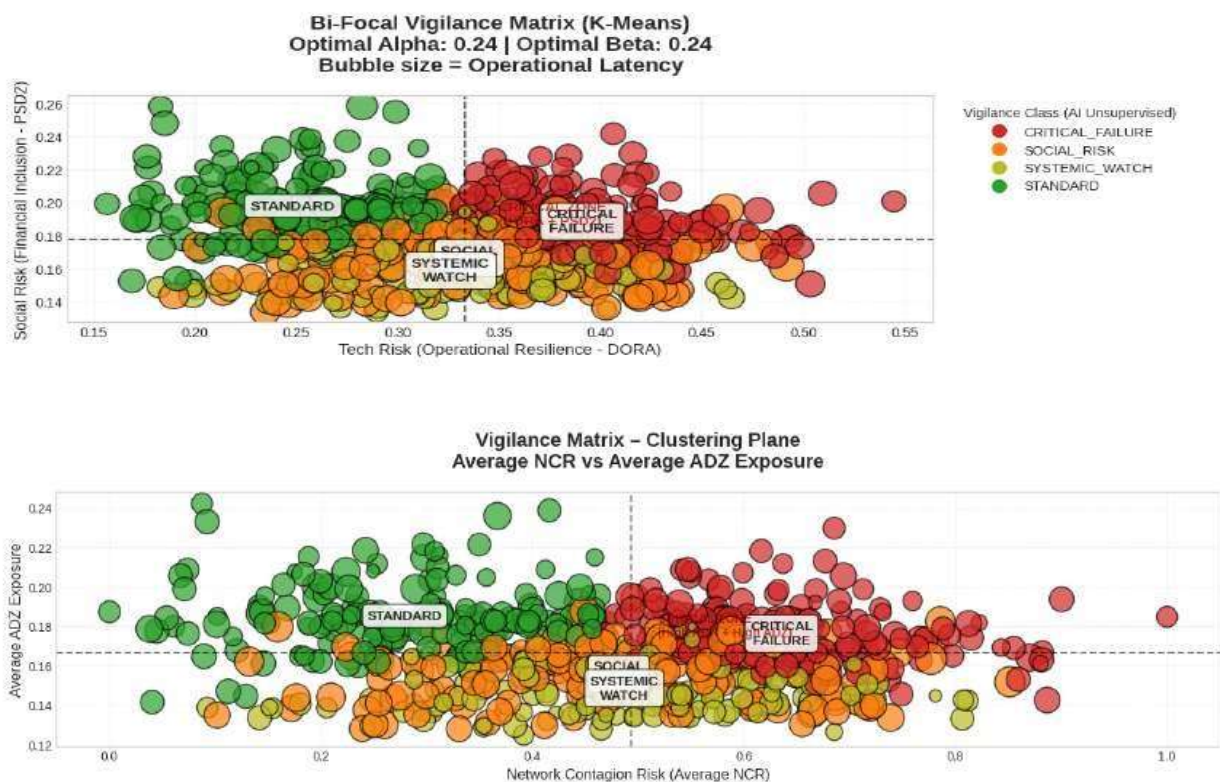
SPS Framework – Supporting Evidence & Key Results

Innovation Data Challenge 2026

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The Bi-Focal Output (Vigilance Matrix)

Objective: Segment Payment Service Providers (PSPs) balancing Technical Contagion Risk (DORA) and Social Exclusion Risk (PSD2) using K-Means clustering.



Key Results:

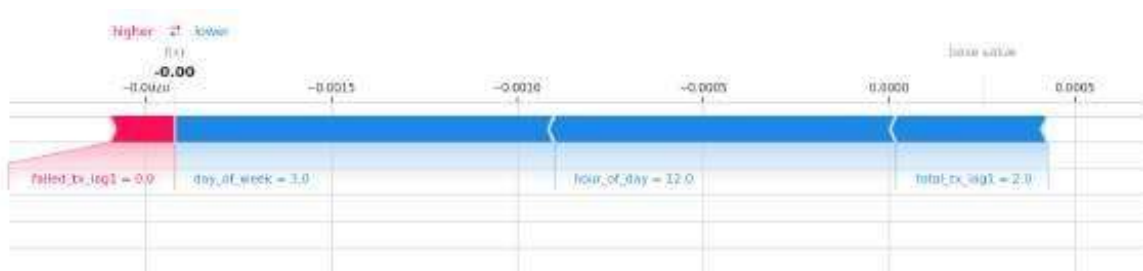
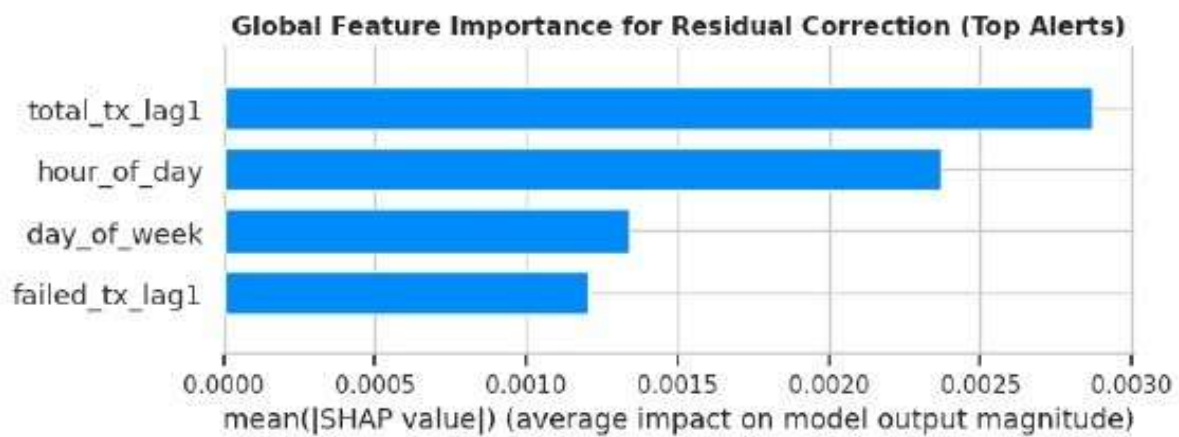
CRITICAL_FAILURE: 201 PSPs require immediate IT inspection.

SOCIAL_RISK: 167 PSPs require inclusion audit (Active Denial Zones).

SYSTEMIC_WATCH: 159 PSPs placed under continuous monitoring.

Predictive Explainability (DORA Art. 9 Compliance)

Objective: Ensure full algorithmic transparency for the Central Bank using SHAP values (Explainable AI), avoiding "black box" decisions.



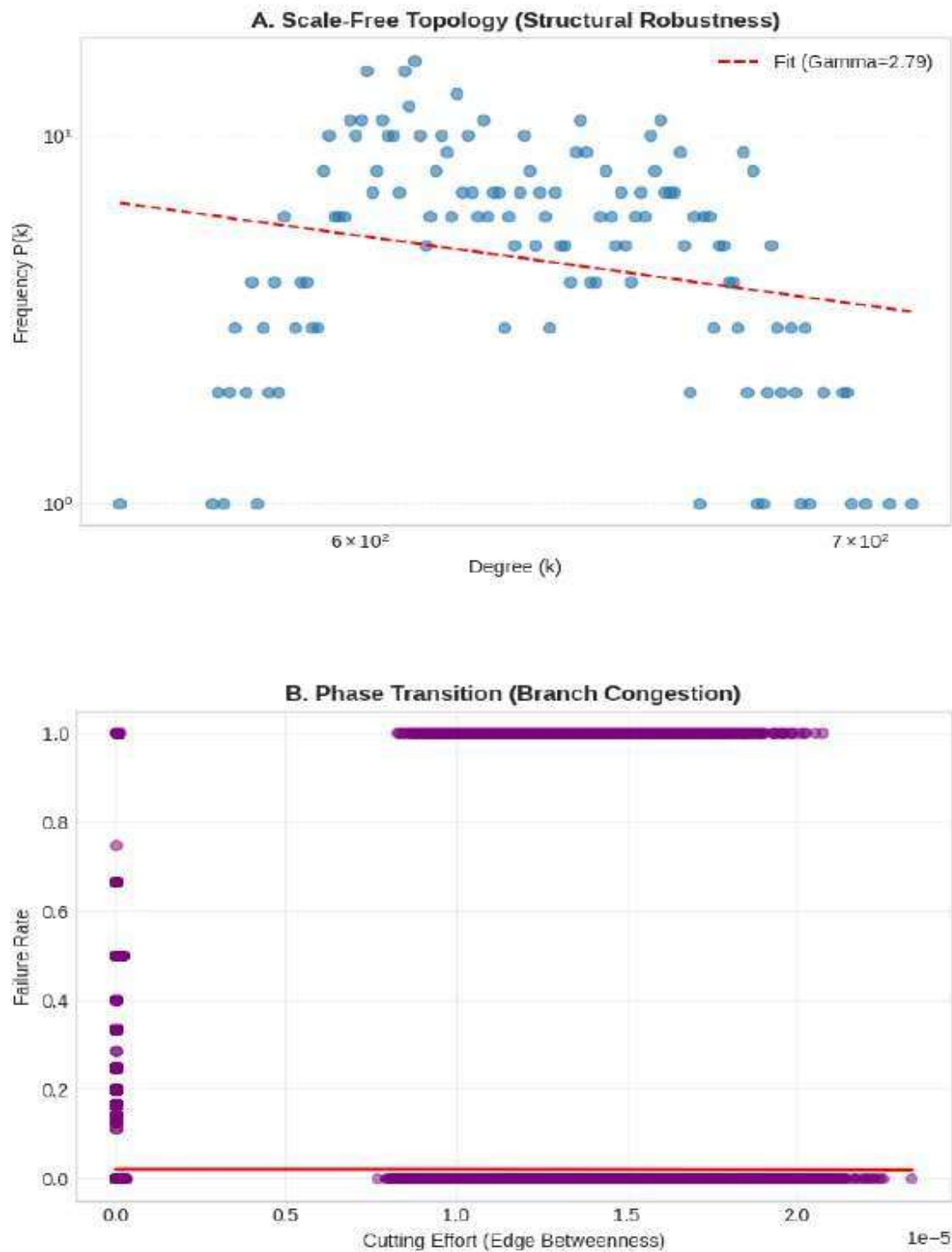
Key Insights:

Prior transaction volume and latency are the dominant drivers of operational failure.

The system isolates the exact causal driver for every alert, satisfying DORA incident management requirements.

Operational Resilience (DORA Stress Test)

Objective: Evaluate the systemic resilience of the retail payment network through Scale-Free Topology analysis and simulated node drop-outs.



Key Insights:

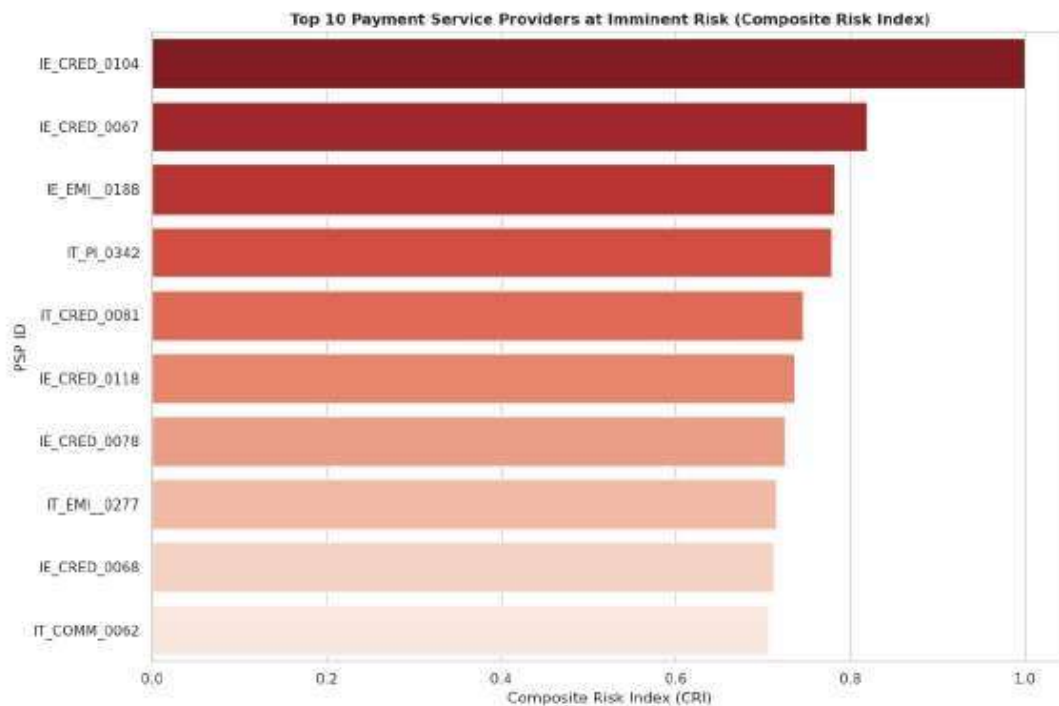
Network exhibits Scale-Free properties ($\gamma \approx 2.79$) with 3 main Tier-1 Super-Hubs. System maintains high topological resilience even under triple-shock scenarios.

Executive Targeting (Top-Risk PSPs)

Objective: Provide supervisors with an actionable, prioritized list of entities most likely to fail in the next 24 hours.

FINAL PREDICTIVE RANKING (Top 10 High-Risk PSPs - Next 24h):

	sender_psp_id	CRI_Score	alert_count	max_z_score
0	IE_CRED_0104	1.000	6	7.027
1	IE_CRED_0067	0.819	5	6.739
2	IE_EMI_0188	0.782	5	8.401
3	IT_PI_0342	0.779	4	6.857
4	IT_CRED_0081	0.746	4	6.942
5	IE_CRED_0118	0.736	4	7.841
6	IE_CRED_0078	0.726	6	3.822
7	IT_EMI_0277	0.715	4	7.037
8	IE_CRED_0068	0.712	5	6.734
9	IT_COMM_0062	0.707	4	6.976



Key Insights:

Predictive 24h risk ranking powered by Negative Binomial GLM + XGBoost residuals.

Empowers Central Banks to allocate inspection resources strictly where systemic risk is concentrated.